Trajectory Tracking Experiments for the WAM-V MPC Follower

Lemniscate path — Bayesian optimization of fifteen control approaches
Open-water Gazebo simulation with ground-truth pose

MPC Experiment Campaign — `molo_wpt_follower/mpc`

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Abstract

This report documents a Bayesian-optimization (BO) and validation campaign to reduce cross-track error (XTE) on a **lemniscate** reference for the WAM-V-16 in open-water Gazebo simulation. Fifteen non-learning guidance and control architectures were compared: each retained its fixed outer-loop law while MPC, ILOS, PID, and guidance parameters were tuned with 28 Gaussian-process BO trials per approach. All final evaluations used a **fixed lemniscate geometry** (77-point figure-8, Dubins step 1.25 m) so that spurious Dubins loops do not distort the reference path. Target: XTE RMSE < 0.1 m.

The best result on the clean reference was **H0_baseline** (`baseline_ilos_velocity` + velocity MPC): RMSE 0.231 m. Velocity-based laws with ILOS guidance ranked highest; spatial MPC, sliding-mode, and several geometric controllers remained unstable. Diagnostic plots now log MPC velocity references (u, v, r) and thruster commands alongside trajectory and XTE. This report presents methodology, control-law mathematics, ranked validation results, and discussion.

Contents

1 Introduction

Accurate path tracking on curved trajectories is challenging for underactuated surface vessels: differential thrust actuates surge and yaw directly, while sway must arise from coupled hydrodynamics. The `molo_wpt_follower/mpc` package implements a cascade architecture in which an outer guidance layer produces velocity references $[u, v, r]$ and an inner model predictive controller (MPC) allocates left/right thrust subject to saturation.

The project goal was **XTE RMSE** < 0.1 m on a lemniscate path. Per-approach Bayesian optimization tuned control parameters; a subsequent **lemniscate validation** campaign (`lemniscate_validation` re-evaluated all BO winners on a shared, verified figure-8 reference (Section 7).

2 System Description

2.1 Vessel and plant model

The WAM-V-16 parameters follow the LINC nominal Fossen model (`boat_model.py`): rigid-body mass, added mass, linear and quadratic damping, and differential thrusters with arm length 0.756 m and maximum thrust 250 N per side. The continuous-time state is

$$\mathbf{z} = [x,\, y,\, \psi,\, u,\, v,\, r]^\top,$$

with world-frame kinematics $\dot{x} = R(\psi)[u, v]^\top$ and body-frame dynamics $M\dot{\nu} = \tau - C(\nu)\nu - D\nu - d_{\mathrm{nl}}(\nu)$.

2.2 Control architectures

Two inner-loop MPC formulations were evaluated:

- **Velocity MPC** (default): regulates $[u, v, r]$ by solving a convex quadratic program (QP) each control step. The solver is **OSQP** (Operator Splitting Quadratic Program) [1]: an open-source first-order method based on the alternating direction method of multipliers (ADMM). OSQP handles linear equality constraints (discretized vessel dynamics) and box constraints on states and thrust; it is invoked from `velocity_mpc.py` via the Python `osqp` interface. References are supplied by `control_law.py` and `velocity_horizon_reference()`.
- **Spatial MPC**: regulates full $\mathbf{z}$ with position weights Q_{xy}, using the same OSQP backend (`spatial_mpc.py`); references come from `horizon_reference()` along the path polyline.

Outer-loop approaches (`control.approach`) are listed in Table 1.

2.3 Simulation environment

Experiments used the `open_water_harner` Gazebo world (no shore collision), WAM-V ground-truth odometry (`use_ground_truth_pose: true`), control rate 10 Hz, evaluation window 55 s with the first 20 s discarded (transient). Runs were executed in Docker (`linc_project`) with ROS 2 Humble; Gazebo was restarted between hypotheses to limit sim-state drift.

3 Methodology

3.1 Scientific workflow

Each hypothesis H_k was documented in `experiments/hypotheses.yaml` with a stated mechanism and prediction. The BO campaign (`tune_lemniscate.py`) implemented:

1. **Hypothesis** — fixed `control.approach` per H_k;

2. **Search** — 28 BO trials with scikit-optimize `gp_minimize`;
3. **Select** — lowest observed trial RMSE;
4. **Validate** — independent Gazebo run with BO-best parameters;
5. **Record** — `tune_result.json`, `best_config.yaml`, logs, plots.

3.2 Performance metric

Cross-track error is published on `/molo_mpc/cross_track_error` and aggregated by `evaluate_mpc.py`:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\mathrm{XTE}_k^2}.$$

Pass criterion: RMSE < 0.1 m.

3.3 Bayesian optimization setup

Campaign `tune_lemniscate_full`: lemniscate trajectory (`scale_m: 7`, closed loop). BO tuned cruise speed, ILOS look-ahead and convergence rate, PID k_p/k_d, guidance gains (curvature, Stanley, SMC, Frenet, backstepping, contouring), MPC horizon, Q_u, Q_v, Q_r, thrust penalty R, and terminal cost scale. Spatial MPC (H11) used a dedicated sub-space for Q_{xy} and Q_ψ. Acquisition: expected improvement (EI).

3.4 Fixed reference path (validation)

Early BO validations could produce **spurious Dubins loops** on the lemniscate when `dubins.step_size_m` was small ($\sim$0.25 m): the planner inserted extra circular arcs between polyline vertices, inflating the reference to thousands of points. The corrected geometry (`reference_path.py`, validated on H9) fixes `resample_step_m`$\approx$0.396 m, `dubins.turning_radius_m`$\approx$5.7 m, and `dubins.step_size_m`=1.25 m, yielding a **77-sample** closed lemniscate for every approach. Final rankings in Section 7 use `run_lemniscate_validation.py`: each hypothesis retains its `best_config.yaml` control parameters but shares this reference geometry.

4 Approaches Under Test

Table 1: Implemented guidance / MPC combinations (hypothesis IDs).

ID	`control.approach`	Inner loop
H0	`baseline_ilos_velocity`	Velocity MPC
H1, H1b, H9, H10	`curvature_ff_ilos`	Velocity MPC
H2	`stanley`	Velocity MPC
H3	`aggressive_ff`	Velocity MPC
H5	`geometric_ff`	Velocity MPC
H8	`pure_pursuit`	Velocity MPC
H11	(n/a)	Spatial MPC
H12	`smc_ilos`	Velocity MPC
H13	`smc_geometric`	Velocity MPC
H14	`frenet_ff`	Velocity MPC
H15	`backstepping`	Velocity MPC
H16	`contouring`	Velocity MPC

Key variants among the top performers (lemniscate validation):

- **H0 baseline** (rank 1): `baseline_ilos_velocity`, conservative cruise ($0.14\,\mathrm{m\,s^{-1}}$); RMSE 0.231 m.
- **H1 curvature FF** (rank 2): `curvature_ff_ilos`; RMSE 0.238 m.
- **H1b tuned FF** (rank 3): same law family with BO-tuned Q_r and κ feedforward; RMSE 0.290 m.
- **H9 slow tight** (rank 4): low cruise, RMSE 0.375 m on the clean reference.

5 Mathematical Formulation of Control Approaches

All velocity-based hypotheses share a **cascade**: an outer guidance law computes references $(u_{\mathrm{cmd}}, v_{\mathrm{ref}}, r_{\mathrm{ref}})$ and a horizon $\nu^{\mathrm{ref}}_{0:N} \in \mathbb{R}^3$; the inner velocity MPC tracks them via OSQP. Spatial MPC (H11) bypasses the outer loop and tracks pose references directly. Implementations are in `control_law.py`, `path_reference.py`, and `velocity_mpc.py` / `spatial_mpc.py`.

5.1 Path geometry and shared quantities

The reference is a resampled polyline $\{(x_i, y_i, \psi_i, \kappa_i)\}_{i=0}^{M-1}$ with curvature $\kappa_i = \mathrm{d}\psi/\mathrm{d}s$ along arc length. At each step the controller finds the closest index $i^\star$ and defines:

$$e_d = \text{signed cross-track error (positive = left of tangent)}, \tag{1}$$

$$\psi_e = \mathrm{wrap}(\psi_{i^\star} - \psi), \qquad u_s = \max(|u_{\mathrm{cmd}}|,\, 0.15). \tag{2}$$

Surge is scheduled from nominal cruise u_c and local curvature:

$$u_{\mathrm{cmd}} = \frac{u_c \cdot \max(0.3,\, e^{-2.2|e_d|})}{1 + \alpha_\kappa |\kappa|},$$

with lower bound $0.12\,\mathrm{m\,s^{-1}}$ unless reverse is allowed. The MPC horizon preview (`velocity_horizon_reference`) propagates along the path with $r_k = \mathrm{clip}(u_k\kappa_k,\, \pm r_{\max})$.

5.2 Integral LOS (ILOS) and heading PID

ILOS (`ILOSFollower`) provides a desired heading ψ_{des} from lateral error y_e in the path frame and a variable look-ahead $\Delta(y_e)$:

$$\dot{\hat{\beta}} = \frac{\gamma\, u\, y_e}{\sqrt{\Delta^2 + (y_e + \Delta\hat{\beta})^2}}, \qquad \psi_{\mathrm{des}} = \psi_{\mathrm{proj}} + \arctan\!\Big(-\hat{\beta} - \frac{y_e}{\Delta}\Big).$$

Heading PID closes the loop on yaw rate:

$$r_{\mathrm{fb}} = k_p \psi_e + k_i \int \psi_e \,\mathrm{d}t + k_d \dot{\psi}_e.$$

5.3 Inner-loop MPC (common to H0–H10, H12–H16)

The Fossen plant is linearized about the current pose $\mathbf{z}$; only the body velocities $\boldsymbol{\nu} = [u, v, r]^\top$ enter the QP. Over horizon N and thrust increments $\mathbf{u}_k = [\Delta T_L, \Delta T_R]^\top$:

$$\begin{aligned}
\min_{\{\boldsymbol{\nu}_k\},\{\mathbf{u}_k\}} \quad & \sum_{k=0}^{N-1} (\boldsymbol{\nu}_k - \boldsymbol{\nu}_k^{\mathrm{ref}})^\top Q (\boldsymbol{\nu}_k - \boldsymbol{\nu}_k^{\mathrm{ref}}) + \mathbf{u}_k^\top R \mathbf{u}_k \\
& + (\boldsymbol{\nu}_N - \boldsymbol{\nu}_N^{\mathrm{ref}})^\top Q_f (\boldsymbol{\nu}_N - \boldsymbol{\nu}_N^{\mathrm{ref}}) \\
\text{s.t.} \quad & \boldsymbol{\nu}_{k+1} = A_d \boldsymbol{\nu}_k + B_d \mathbf{u}_k, \quad \boldsymbol{\nu}_0 = \boldsymbol{\nu}_{\mathrm{meas}}, \\
& \boldsymbol{\nu}_{\min} \le \boldsymbol{\nu}_k \le \boldsymbol{\nu}_{\max}, \quad \mathbf{u}_{\min} \le \mathbf{u}_k \le \mathbf{u}_{\max}.
\end{aligned} \tag{3}$$

Applied thrust is $\mathbf{T} = \mathbf{u}_0 + \mathbf{T}_{\mathrm{ff}}(\boldsymbol{\nu}_0^{\mathrm{ref}})$.

5.4 Outer-loop guidance laws

Table 2 summarizes each `control.approach`; g_κ denotes `kappa_ff_gain`, k_{cte} `stanley_k`, and $\text{sat}(s) = \tanh(s/\phi)$ the SMC boundary layer.

Table 2: Mathematical structure of each guidance law (references passed to velocity MPC).

Approach	Law
`baseline_ilos_velocity`	ψ_{des} from ILOS; $r_{\text{ref}} = \text{clip}(r_{\text{fb}}, \pm r_{\max})$; $\nu_k^{\text{ref}} \equiv [u_{\text{cmd}}, 0, r_{\text{ref}}]^\top$.
`curvature_ff_ilos`	Curvature feedforward $r_{\text{ff}} = g_\kappa\, u_{\text{cmd}}\, \kappa$; $r_{\text{ref}} = \text{clip}(r_{\text{ff}} + r_{\text{fb}}, \pm r_{\max})$; optional $v_{\text{ref}} = -k_{\text{lat}} e_d$; horizon blends path-preview $r = u\kappa$ with $0.3\, r_{\text{fb}}$.
`stanley`	Nonlinear heading law $\psi^\star = \psi_{i^\star} + \text{atan2}(k_{\text{cte}} e_d,\, u_s)$; $r_{\text{ref}} = \text{clip}(0.7\, u_{\text{cmd}}\kappa + 0.3\, \psi^\star - \psi/\Delta t)$.
`aggressive_ff`	$r_{\text{ref}} = \text{clip}(u_{\text{cmd}}\kappa + 2k_{\text{cte}} e_d)$ or $r_{\text{ff}} + r_{\text{fb}}$; stronger $v_{\text{ref}} = -0.8 e_d$.
`geometric_ff`	Path-tangent tracking: $r_{\text{ref}} = u_{\text{cmd}}\kappa + k_{\text{cte}} e_d/u_s + \frac{1}{2} r_{\text{fb}}$; $v_{\text{ref}} = -e_d$; horizon yaw rate augmented by $k_{\text{cte}} e_d/u_s$.
`pure_pursuit`	Look-ahead point $\mathbf{p}_{\text{la}}$ at fixed index offset; $\psi_{\text{pp}} = \text{atan2}(y_{\text{la}} - y,\, x_{\text{la}} - x)$, $r_{\text{pp}} = (\psi_{\text{pp}} - \psi)/\Delta t$; $r_{\text{ref}} = \text{clip}(0.5 r_{\text{pp}} + 0.5 r_{\text{ff}} + 0.25 r_{\text{fb}})$.
`smc_ilos` / `smc_geometric`	Sliding surfaces $s_y = e_d + \lambda_i \int e_d\, \text{d}t$, $s_\psi = \text{wrap}(\psi_{\text{des}} - \psi)$; $r_{\text{ref}} = g_\kappa u_{\text{cmd}} \kappa + k_r\, \text{sat}(s_\psi)$, $v_{\text{ref}} = -k_v\, \text{sat}(s_y)$. ILOS variant uses ψ_{des} from ILOS; geometric variant uses integral-LOS heading on path tangent.
`frenet_ff`	Frenet-style lateral regulation on polyline: $r_{\text{ref}} = g_\kappa u_{\text{cmd}}\kappa + k_y e_d/u_s + k_\psi \psi_e$; $v_{\text{ref}} = -0.8 k_y e_d$.
`backstepping`	Virtual heading $\alpha = \text{wrap}\big(\psi_{i^\star} - \text{atan2}(k_1 e_d,\, \delta_{\text{los}} + u_s)\big)$, $z_2 = \text{wrap}(\psi - \alpha)$; $r_{\text{ref}} = -k_2 z_2 - z_2/\Delta t + g_\kappa u_{\text{cmd}}\kappa$.
`contouring`	Contouring error rate $\dot{e}_d \approx u_{\text{cmd}} \sin(\psi - \psi_{i^\star})$; $r_{\text{ref}} = g_\kappa u_{\text{cmd}}\kappa + k_d e_d + k_{dd}\dot{e}_d$; $v_{\text{ref}} = -0.9 k_d e_d$.

5.5 Spatial MPC (H11)

Spatial MPC regulates the full state $\mathbf{z} = [x, y, \psi, u, v, r]^\top$ against polyline pose references $\mathbf{z}_k^{\text{ref}} = [x_k, y_k, \psi_k, u_{\text{ref}}, 0, 0]^\top$ from `horizon_reference()` (no curvature feedforward in r). The cost penalizes (x, y) with weights Q_{xy}, heading with Q_ψ, and velocities with Q_u, Q_v, Q_r: same QP structure as above but $n_x = 6$ and $\mathbf{z}_0 = \mathbf{z}_{\text{meas}}$. Heading references are unwrapped relative to current ψ before each solve.

6 Bayesian Optimization Campaign

Fifteen hypotheses were tuned independently in Gazebo (`experiments/tune_lemniscate.py`, output `tune_lemniscate_full/`). For each H_k the outer-loop law `control.approach` was held fixed; a Gaussian-process surrogate over $\sim$26 continuous parameters (cruise speed, ILOS, PID, guidance gains, MPC weights; H11 used a spatial sub-space) was minimized with scikit-optimize `gp_minimize` and expected-improvement acquisition. Each hypothesis received **28 BO trials**; after each trial the simulator was restarted, XTE RMSE was measured over 55 s (first 20 s discarded), and the parameter vector with lowest observed RMSE was written to `best_config.yaml`.

Path geometry is now **fixed** for all trials via `reference_path.py` (Section 3): BO optimizes tracking performance, not Dubins/resample settings. The rankings reported below do *not* use the original per-trial reference paths; instead, Section 7 re-validates every BO winner on the shared 77-point lemniscate so that controller quality is compared fairly on identical geometry.

7 Results: Lemniscate Validation (Fixed Reference)

Following the BO campaign, `run_lemniscate_validation.py` loaded each `best_config.yaml`, replaced only the path/Dubins blocks with the fixed reference from `reference_path.yaml`, and ran one independent Gazebo evaluation per hypothesis (results in `lemniscate_validation/`). Table 3 and Figure 1 summarize the **six best** approaches on this common reference. Lower-ranked hypotheses (H8, H2, H12–H16) achieved RMSE from 4 m to 122 m and are omitted from the bar chart for readability.

Table 3: Top six approaches: lemniscate validation RMSE with fixed reference path. Data: `lemniscate_validation/validation_summary.json`. Target < 0.1 m.

Rank	Hypothesis	RMSE [m]
1	H0_baseline	**0.231**
2	H1_curvature_ff	0.238
3	H1b_tuned_ff	0.290
4	H9_slow_tight	0.375
5	H10_lateral_vel	0.507
6	H5_geometric_ff	0.931

Ranks 7–15: H8 (23.3 m) through H11 spatial (121 m).

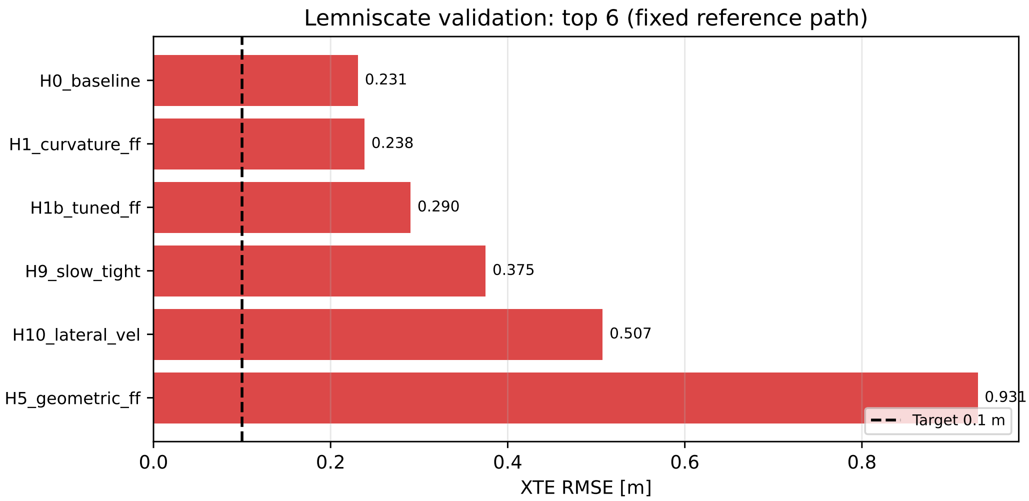


Figure 1: Top six approaches: XTE RMSE on the fixed lemniscate reference (`lemniscate_validation`). Dashed line: 0.1 m target.

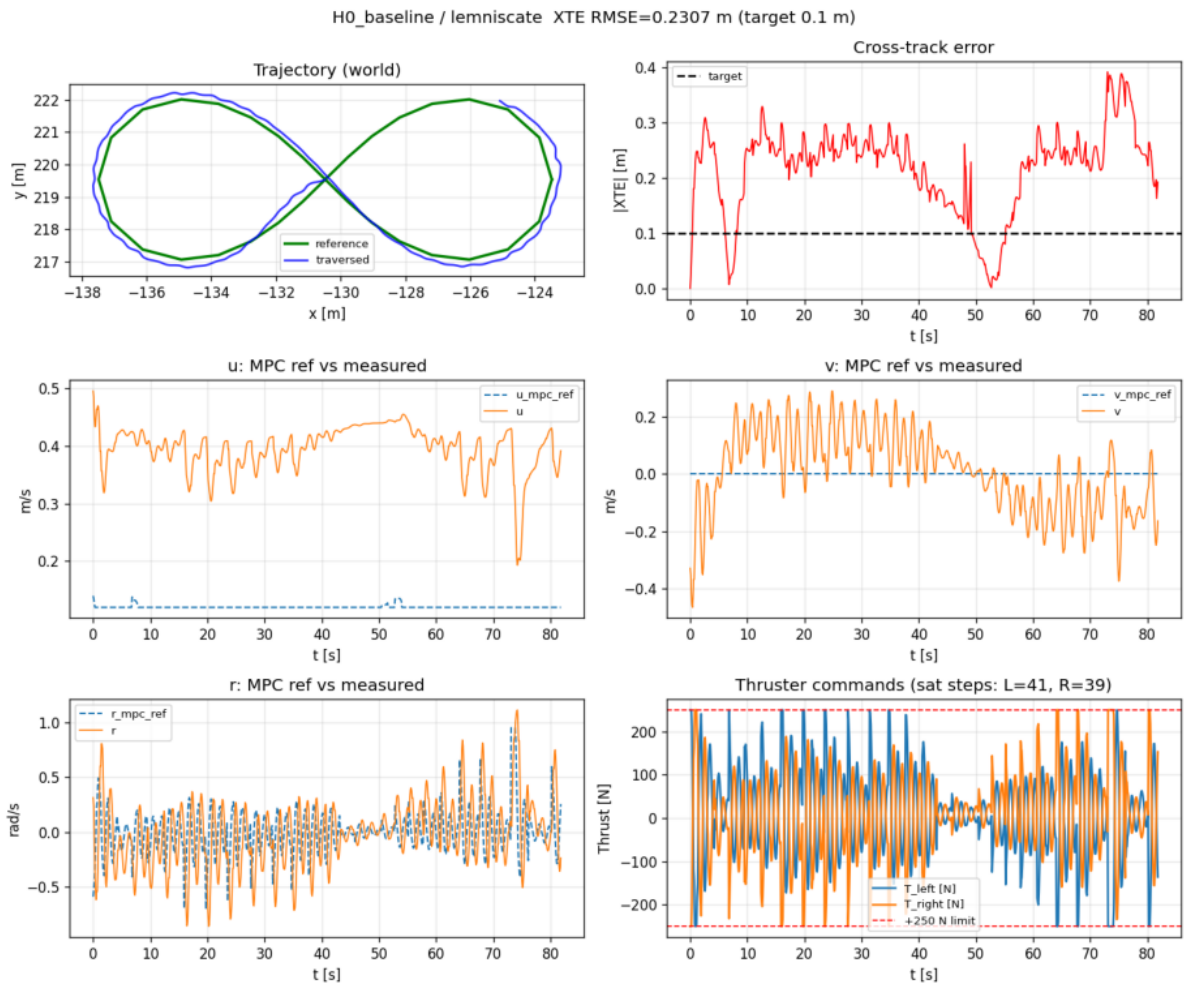
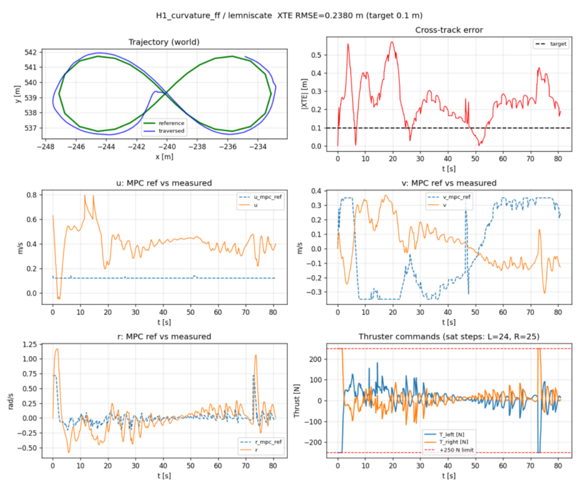
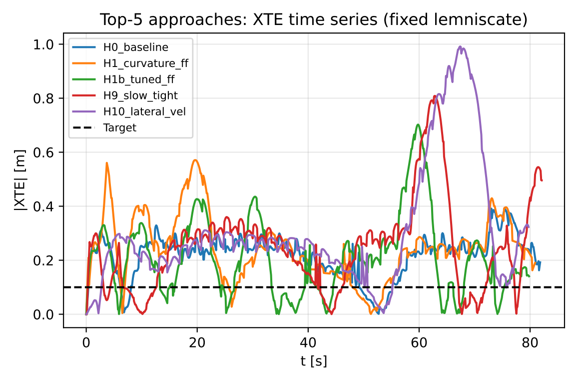


Figure 2: Best approach H0_baseline on the fixed lemniscate: trajectory, XTE, velocity MPC tracking (u, v, r vs. `nu_horizon[0]`), and thruster commands with saturation limits ($\pm 250\,\mathrm{N}$).



(a) H1_curvature_ff (rank 2)



(b) Top-5 XTE time series (H5 omitted from top-6)

Figure 3: Runner-up diagnostics and cross-approach XTE comparison.

Key findings:

- **Winner**: H0_baseline — `baseline_ilos_velocity` + velocity MPC, RMSE 0.231 m (closest to target among all fifteen approaches).
- **Runner-up**: H1_curvature_ff at 0.238 m; H1b tuned FF at 0.290 m.
- **H9 slow tight**: 0.375 m on the correct reference (previously 0.109 m on a path corrupted by small Dubins step size).
- **Diagnostics**: extended logging shows MPC tracks ν_0^{ref} while guidance scalars (`u_cmd`) can differ; thruster saturation is visible on tight curve segments.
- **Target**: none of the fifteen approaches met 0.1 m.

8 Discussion

8.1 Why velocity MPC + curvature FF succeeded

The top stack separates **feasible** inner-loop tracking ($[u, v, r]$ via OSQP) from **geometric** outer-loop planning (ILOS + $r = u\kappa$ preview). On this validation run the conservative H0 baseline edged out curvature-FF variants; H1b's BO-tuned Q_r and κ feedforward did not improve over plain ILOS once the reference path was corrected. Reduced cruise speed (H9) alone was insufficient.

8.2 Why lower-ranked approaches failed after BO

Spatial MPC (H11, 122 m): path-index jumps on closed curves, large Q_{xy} on an underactuated plant, and zero r_{ref} in the spatial preview caused runaway thrust.

SMC, Frenet, backstepping, contouring (H12–H16): aggressive lateral and heading gains produced oscillation; BO trial minima were not reproducible on validation.

Stanley, pure pursuit (H2, H8): unsuitable for closed lemniscate without structural changes (lookahead / monotonic arc-length indexing).

8.3 Diagnostic logging

`log.csv` records measured $[u, v, r]$, MPC references `u_mpc_ref`, `v_mpc_ref`, `r_mpc_ref` (from `nu_horizon[0]`), and pre-saturation thrust in newtons. Per-hypothesis `plots.png` overlays MPC refs against measured velocities and marks thruster saturation at ± 250 N. Guidance scalars (`u_cmd`, `r_ref`) are retained for outer-loop analysis but are not the MPC setpoint.

9 Conclusions

1. **Best approach**: **H0_baseline** — `baseline_ilos_velocity` + velocity MPC (OSQP); validation RMSE 0.231 m on the fixed lemniscate reference.

2. **Goal**: RMSE < 0.1 m not achieved; best margin $\approx$0.131 m.

3. **Reference path matters**: rankings changed substantially once all approaches shared the same 77-point figure-8; Dubins step size must not be BO-tuned on closed curves.

4. **Not recommended without redesign**: spatial MPC, pure pursuit, contouring/backstepping on closed tight curves in this sim stack.

10 Future Work

- **Contouring / Frenet NMPC**: penalize $e_d, \dot{e}_d$ with arc-length state.

- **Spatial MPC v2**: monotonic arc-length index, $r_{\mathrm{ref}} = u\kappa$ preview.

- **Hybrid**: retain H9 outer loop; add light position feedback in cost.
- **Validation**: multiple Gazebo seeds per BO winner; report mean ± std.

A Reproduction

```
cd molo_wpt_follower/mpc
python3 experiments/tune_lemniscate.py --n-calls 28
python3 experiments/run_lemniscate_validation.py
python3 experiments/plot_results.py experiments/results/lemniscate_validation --validation
python3 report/generate_figures.py
cd report && make pdf
```

BO configs: `experiments/results/tune_lemniscate_full/`
Validation: `experiments/results/lemniscate_validation/`

B Hypothesis registry

Full statements in `experiments/hypotheses.yaml` and `experiments/approaches.yaml`.

References

[1] P. Stellato, G. Banjac, G. Goulart, A. Bemporad, and S. Boyd, "OSQP: an operator splitting solver for quadratic programs," *Mathematical Programming Computation*, 2020.